

Analysis for Systematic Dimensions of Metal Hall Coefficients

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1 Introduction and Experimental Overview

When a conducting slab carrying longitudinal electric current I_x is exposed to a magnetic field B_z perpendicular to the slab, moving charge carriers will experience a transverse Lorentz force:

$$\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B}) \quad (1)$$

This force deflects charges towards one edge of the conductor until the accumulation of charge begins generating an opposing transverse electric field E_y which is also known as the Hall field. Once at steady state, the net force in the transverse direction vanishes, $E_y = v_x B_z$.

For a rectangular sample of thickness t , length l , and width w , the current density in the longitudinal direction is $J_x = nqv_x = I_x/(wt)$, where n is the charge carrier number density and q is the carrier charge. The resulting transverse potential difference (Hall Voltage), $V_H = -E_y w$, is expressed as:

$$V_H = - \left(\frac{1}{nq} \right) \frac{I_x B_z}{t} \quad (2)$$

Where the term in the parenthesis is the Hall coefficient, R_H :

$$R_H = \frac{1}{nq} = - \frac{V_H t}{I_x B_z} \quad (3)$$

To get the true Hall voltage V_H from geometric offset voltages $V_y(B_z = 0)$ caused by contact misalignment, our measurements were acquired across two 180° probe orientations relative to the magnetic field poles. The potential difference between the orientations yielded twice the magnitude of the Hall voltage:

$$2V_H = V_y(\text{orientation 1}) - V_y(\text{orientation 2}) \quad (4)$$

Our calibrated magnetic field of $B_z = 0.133467 \pm 0.002041$ T was used to perform linear regression on $2V_H$ as a function of excitation current I_x to extract the response slope $m = 2V_H/I_x$. The experimental Hall coefficient is calculated via:

$$R_H = - \frac{t}{2B_z} \left(\frac{2V_H}{I_x} \right) \quad (5)$$

2 Experimental Results

Table 1 shows the sample parameters, calculated experimental Hall coefficients, conductivities ($\sigma = l/(Rwt)$), and carrier densities.

Table 1: Experimental Transport Properties

Material	Thickness t (m)	$R_{H, \text{exp}}$ ($\text{m}^3/\text{A} \cdot \text{s}$)	σ (S/m)	Carrier Density n (m^{-3})
Gold (Au)	$(1.37 \pm 0.16) \times 10^{-7}$	$(-5.57 \pm 0.67) \times 10^{-11}$	$(2.28 \pm 0.27) \times 10^7$	$(1.12 \pm 0.14) \times 10^{29}$
Aluminum (Al)	$(2.34 \pm 0.17) \times 10^{-7}$	$(-2.67 \pm 0.21) \times 10^{-11}$	$(1.177 \pm 0.089) \times 10^7$	$(2.34 \pm 0.18) \times 10^{29}$
InAs	$(1.26 \pm 0.02) \times 10^{-4}$	$(-1.672 \pm 0.037) \times 10^{-4}$	$(1.36 \pm 0.31) \times 10^4$	$(3.734 \pm 0.082) \times 10^{22}$

Table 2: Fitted Voltage Slopes and Charge Carrier Mobilities

Material	Slope $m = \frac{2V_H}{I_x}$ (V/A)	Carrier Mobility μ ($\text{m}^2/\text{V} \cdot \text{s}$)
Gold (Au)	$(1.086 \pm 0.029) \times 10^{-4}$	$(1.27 \pm 0.21) \times 10^{-3}$
Aluminum (Al)	$(3.045 \pm 0.060) \times 10^{-5}$	$(3.14 \pm 0.34) \times 10^{-4}$
InAs	0.35414 ± 0.00017	2.27 ± 0.53

Table 2 summarizes the voltage-current response slopes ($m = 2V_H/I_x$) along with the calculated Hall mobilities ($\mu = \sigma|R_H|$).

The statistical uncertainty in R_H for the metallic probes is overwhelmed by the thickness measurement:

- **Gold probe error:** The magnetic field B_z contributes 1.604%, the thickness t contributes 93.55%, and slope m contributes 4.846%.
- **Aluminum probe error:** The magnetic field B_z contributes 3.959%, the thickness t contributes 89.363%, and slope m contributes 6.678%.

3 Al and Au: Experimental and Literature Values Comparison

3.1 Dimensional Discrepancies

The experimental Hall coefficients obtained for both Al and Au are lower in magnitude than the accepted room-temperature literature values given by Hurd [1]:

- **Aluminum:** $R_{H, \text{exp}} = (-2.67 \pm 0.21) \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ vs. Hurd’s literature values of $-3.44 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ (287.6 K) and $-3.50 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ (291.7 K).
- **Gold:** $R_{H, \text{exp}} = (-5.57 \pm 0.67) \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ vs. Hurd’s literature values of $-7.05 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ (290 K) and $-7.36 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$ (RT).

Assigning inaccuracies in the sample dimensions for this difference in values, we should examine how the parameters are used in equation (3):

1. **Sample Thickness (t):** Since $R_H \propto t$, an underestimation of the sample thickness will reduce the calculated value of $|R_H|$. Aligning the experimental magnitudes with those from Hurd’s literature we increase thickness values:
 - For **Aluminum**, increasing t from $2.34 \times 10^{-7} \text{ m}$ to $3.02 \times 10^{-7} \text{ m}$ or $3.07 \times 10^{-7} \text{ m}$ (an increase of $\approx 29\text{--}31\%$) brings $|R_{H, \text{exp}}|$ to $3.44\text{--}3.50 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$.
 - For **Gold**, increasing t from $1.37 \times 10^{-7} \text{ m}$ to $1.73 \times 10^{-7} \text{ m}$ or $1.81 \times 10^{-7} \text{ m}$ (an increase of $\approx 26\text{--}32\%$) brings $|R_{H, \text{exp}}|$ to $7.05\text{--}7.36 \times 10^{-11} \text{ m}^3/\text{A} \cdot \text{s}$.

2. **Sample Width (w) and Length (l):** The width and length of the sample don't appear in equation (3) for R_H . So changing w or l will impact the sample resistance R and conductivity σ , but it will have no effect on the measured Hall coefficient R_H .

3.2 Evaluation of Consistency

Comparing the aluminum and gold metals shows that there is a consistent underestimation of Hurd's accepted literature values for $|R_H|$ by approximately 20–25%:

$$\frac{|R_{H, \text{exp}}|}{|R_{H, \text{lit}}|} \approx 0.763 - 0.776 \quad (\text{Aluminum}), \quad \frac{|R_{H, \text{exp}}|}{|R_{H, \text{lit}}|} \approx 0.757 - 0.790 \quad (\text{Gold}) \quad (6)$$

The ratio shows that there is high experimental consistency between the two metallic samples and that the discrepancy is not due to random experimental fluctuations, but a systematic offset with the experiment.

Since thickness t for both metallic probes was measured using the same calibration standard, a systematic error in the thickness measurement setup would uniformly underestimate thickness t for both probes. Additionally, variance propagation showed that uncertainty in the thickness t dominated the error introduced (89.363% for Al and 93.55% for Au). Therefore, the experimental results are indeed highly consistent with the expectations for the experimental setup, with the offset being explained by systematic dimensional calibration of the thickness of the films.

References

- [1] C. M. Hurd, *The Hall Effect in Metals and Alloys* (Plenum Press, New York, 1972), pp. 240–253.